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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ARTIFICIAL INTELLIGENCE

PROJECT



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HYBRID RETRIEVAL-AUGMENTED GENERATION FOR ENHANCED LLM-BASED QUESTION ANSWERING

Base Paper: "Hybrid Retrieval-Augmented Generation Approach for LLMs Query Response Enhancement"

1. OBJECTIVES OF THE PROJECT

The main objective of this project is to implement the Hybrid Retrieval-Augmented Generation approach proposed in the selected IEEE base paper. The system combines Sentence-Window and Parent-Child retrieval with a re-ranking stage so that an LLM receives both fine-grained relevant information and broader contextual information.

- Implement a baseline Naive RAG pipeline for comparison.
- Implement Sentence-Window Retrieval for fine-grained retrieval from smaller text units while retaining surrounding context.
- Implement Parent-Child Retrieval by matching child chunks and using their corresponding parent chunks to preserve broader context.
- Combine the outputs of the two retrieval methods and re-rank the candidate contexts.
- Provide the selected top-K contexts to an LLM and compare the approaches using correctness, relevancy, and faithfulness.

2. PROBLEM STATEMENT

Large Language Models can generate coherent natural-language responses, but they may not contain the specific external knowledge required to answer a query accurately. Retrieval-Augmented Generation addresses this limitation by retrieving relevant information from external knowledge sources and supplying it to the LLM as context.

The quality of a RAG response depends strongly on the quality and completeness of retrieved context. Very fine-grained retrieval can identify highly relevant information but may lose surrounding meaning, while broader retrieval can preserve context but may include less relevant information. The project therefore addresses the problem of combining these complementary retrieval strategies and re-ranking their results so that the LLM receives context that is both relevant and sufficiently comprehensive.

3. ABSTRACT

Large Language Models (LLMs) have demonstrated remarkable capabilities in understanding and generating natural language. However, LLMs may produce inaccurate or incomplete responses when the required information is not available in their training knowledge. Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving relevant information from external documents and providing it as contextual information to the LLM before generating a response. However, conventional RAG approaches may retrieve incomplete or less relevant context, which can affect the quality and reliability of the generated answers.

This mini project aims to implement a Hybrid Retrieval-Augmented Generation (RAG) framework based on the proposed approach in the selected IEEE base paper, “Hybrid Retrieval-Augmented Generation Approach for LLMs Query Response Enhancement.” The proposed system combines Sentence-Window Retrieval and Parent-Child Retrieval to improve both the precision and contextual

completeness of retrieved information. Sentence-Window Retrieval identifies highly relevant sentences and incorporates their surrounding context, while Parent-Child Retrieval identifies relevant smaller chunks and associates them with their corresponding larger parent context. The results obtained from these retrieval strategies are then combined and re-ranked to select the most relevant context before passing it to the LLM.

The system will be implemented as a comparative framework consisting of Naive RAG, Sentence-Window RAG, Parent-Child RAG, and the proposed Hybrid RAG approach. User queries will be processed against a collection of documents, relevant information will be retrieved using the respective strategies, and the selected context will be provided to an LLM for response generation. The performance of the different approaches will be evaluated using measures such as correctness, relevancy, and faithfulness of the generated responses.

The proposed system demonstrates the application of Generative AI, Large Language Models, Natural Language Processing, Information Retrieval, and knowledge-based question answering in a unified AI framework.

4. OVERALL ARCHITECTURE DIAGRAM

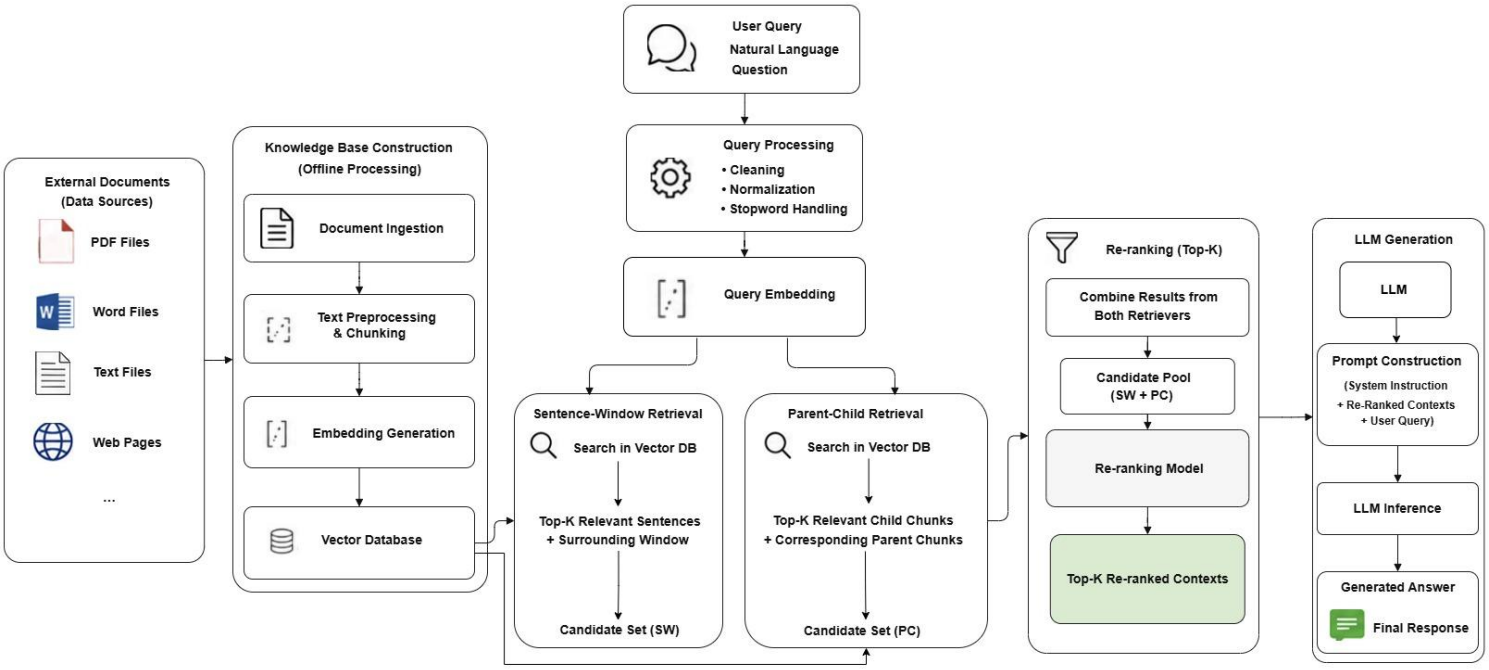


Figure 1. Overall architecture of the Hybrid RAG system.

The overall architecture follows the proposed method in the base paper. The user query activates two retrieval mechanisms in parallel: Sentence-Window and Parent-Child. Each retrieval mechanism produces top-K candidate contexts. These candidates are combined and passed to the re-ranking stage, which selects the highest-scoring contexts before they are supplied to the LLM for answer generation.

5. DETAILED DESIGN DIAGRAM – MODULE-WISE

For the detailed design, the system is divided into six core functional modules. The division is intentionally limited to the major processing stages of the proposed Hybrid RAG pipeline; baseline comparison and evaluation are treated as supporting activities rather than separate runtime modules.

5.1 Module 1 – Knowledge Base Preparation

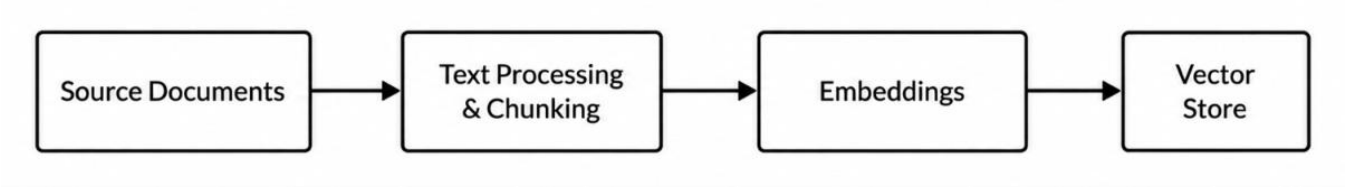


Figure 2. Detailed design of Module 1 – Knowledge Base Preparation.

This module prepares the external knowledge used by the RAG system. Source documents are processed into manageable text units, and the resulting representations are converted into embeddings for retrieval. The vector store maintains the searchable representation of the document knowledge base.

Output: A searchable knowledge base containing the representations required by the retrieval mechanisms.

5.2 Module 2 – Query Processing

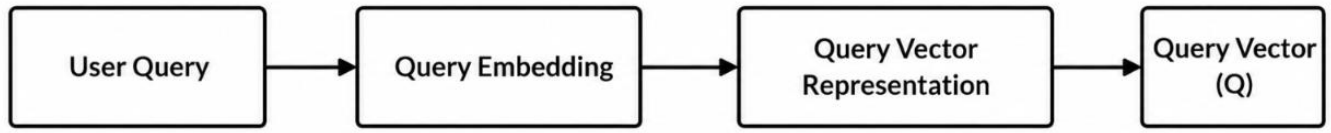


Figure 3. Detailed design of Module 2 – Query Processing.

The user supplies a natural-language query. The query is transformed into a vector representation using an embedding model so that it can be compared with the stored representations in the retrieval stage. This is the common query-processing path used before the two retrieval mechanisms operate.

Output: Query vector representation used by both Sentence-Window and Parent-Child retrieval.

5.3 Module 3 – Sentence-Window Retrieval

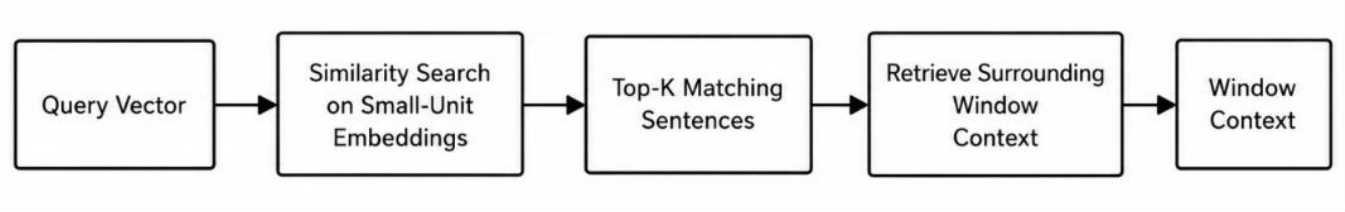


Figure 4. Detailed design of Module 3 – Sentence-Window Retrieval.

Sentence-Window Retrieval focuses on smaller units of text within a larger context. The query is matched against embeddings derived from small text units such as sentences or sentence fragments. The most relevant units are selected as top-K matches, while the surrounding window provides additional context. This gives the retrieval process fine-grained relevance without discarding nearby information.

Output: Top-K sentence-level matches with their surrounding contextual window.

5.4 Module 4 – Parent-Child Retrieval

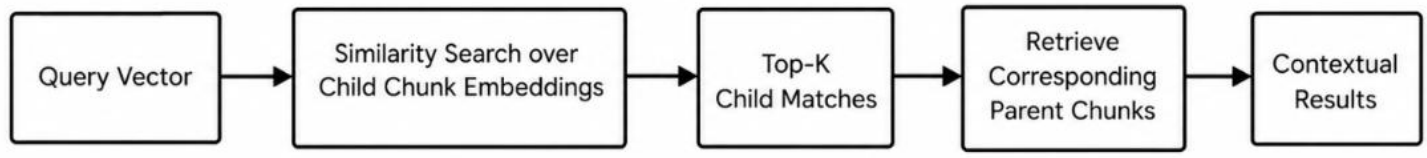


Figure 5. Detailed design of Module 4 – Parent-Child Retrieval.

Parent-Child Retrieval represents information at two levels. A child chunk is a smaller segment that can match the query specifically, while a parent chunk is a broader section containing one or more related child chunks. The retrieval process identifies relevant child chunks and associates them with their corresponding parent chunks so that detailed information and broader context are available together.

Output: Top-K child matches together with their corresponding parent context.

5.5 Module 5 – Hybrid Fusion and Re-ranking

This module forms the central contribution of the proposed method. The top-K results from Sentence-Window and Parent-Child retrieval are combined into a single candidate set. The combined results are then re-ranked to refine relevance and contextual alignment. In the base paper's experimental setting, the re-ranking stage selects the three contexts with the highest scores and sends them to the LLM and it is shown in Figure 6.

Output: Final top-K re-ranked contexts for answer generation.

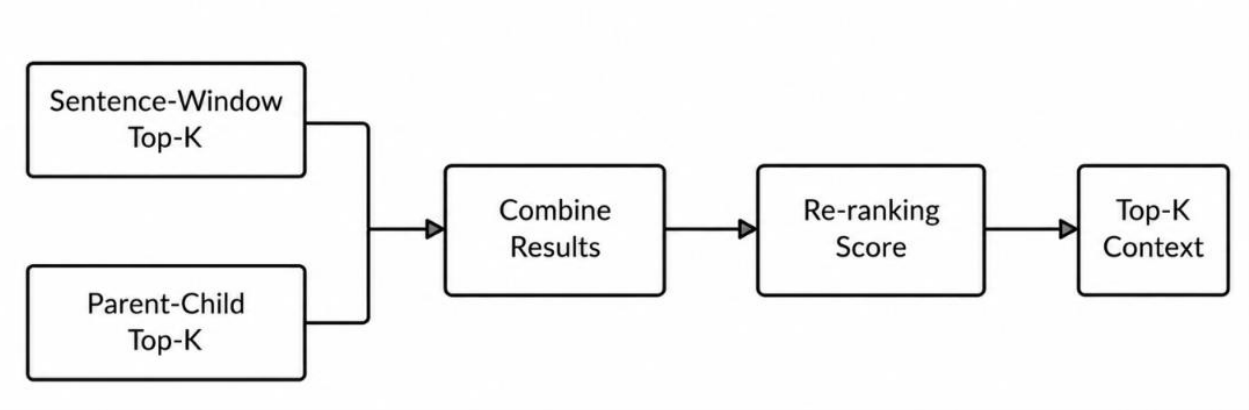


Figure 6. Detailed design of Module 5 – Hybrid Fusion and Re-ranking.

5.6 Module 6 – LLM Response Generation

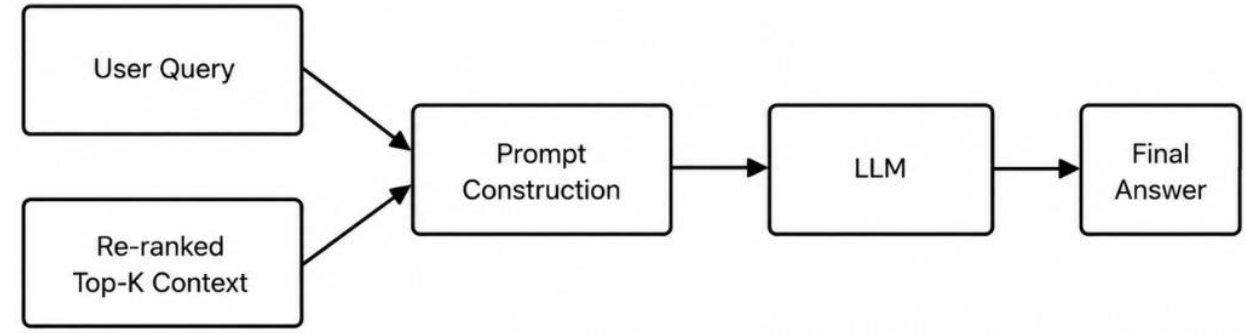


Figure 7. Detailed design of Module 6 – LLM Response Generation.

The original user query and the re-ranked top-K contexts are assembled into the input prompt for the LLM. The base paper uses GPT-3.5 across the compared RAG methods. The LLM uses the retrieved external information as contextual input and generates the final natural-language response.

Output: Generated answer supported by the retrieved context.

6. COMPARATIVE FRAMEWORK

The project reproduces the paper's comparison between the baseline and advanced retrieval configurations. The proposed Hybrid RAG is evaluated against Naive RAG, Sentence-Window, and Parent-Child approaches using the same general response-generation setting.

The project considers four different RAG approaches for comparison. **Naive RAG** uses the user query to retrieve relevant information from the vector store, which is then provided to the LLM for generating the final answer. **Sentence-Window RAG** performs retrieval using smaller text units and includes the surrounding sentences to provide additional context for the response. **Parent-Child RAG** retrieves relevant child chunks and then uses their corresponding parent chunks to provide broader contextual information. Finally, the **proposed Hybrid RAG approach** combines the results obtained from both Sentence-Window and Parent-Child retrieval. These combined results are re-ranked based on their relevance, and the most suitable top-K contexts are provided to the LLM for generating the final response.

7. EVALUATION CRITERIA

The base paper evaluates the response quality using correctness, relevancy, and faithfulness. These measures will be used as the project's primary evaluation criteria.

- Correctness: Evaluates whether the generated response is accurate and relevant to the query and reference answer. The paper uses a 1–5 score.
- Relevancy: Evaluates whether the generated response is relevant to the query and aligned with the retrieved context. The paper uses a score of 0 to 1.
- Faithfulness: Evaluates whether the generated response is supported by the supplied context. The paper uses a score of 0 to 1.

8. EXPECTED OUTCOME

The expected outcome is a working Hybrid RAG implementation that follows the retrieval structure of the selected base paper. The project should demonstrate the individual behavior of Sentence-Window and Parent-Child retrieval and show how combining their results with re-ranking can improve the context supplied to an LLM. The final implementation will compare the four retrieval configurations and report the measured evaluation scores.

9. REFERENCE

P. Omrani, A. Hosseini, K. Hooshanfar, Z. Ebrahimian, R. Toosi, and M. A. Akhaee, "Hybrid Retrieval-Augmented Generation Approach for LLMs Query Response Enhancement," 2024 10th International Conference on Web Research (ICWR), 2024.